PRODUCT REQUIREMENTS DOCUMENT

NetraJyoti MVP - Bengali-first, non-diagnostic eye-care navigation and referral support

Field	Value
Product	NetraJyoti AI
Document status	Capstone MVP requirements baseline, aligned to current live implementation
Primary market	Rural Bengali-speaking communities in West Bengal
Product boundary	Guidance and referral support; not a diagnostic or emergency service
Version	5.2 / 3 September 2026

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Decision summary: The live MVP selects URGENT, ROUTINE or HUMAN SUPPORT through Java deterministic routing after PostgreSQL criterion retrieval. Optional local Ollama runs only after the route is fixed, using PostgreSQL route-guidance records to create a short Bengali explanation. Ollama cannot select or change the route. The current database records are demo-simulated and not clinically approved; Java fallback guidance remains available for every failure.

1. Overview

NetraJyoti is a Bengali voice-first support system for rural West Bengal. It helps people describe an eye-health concern, understand a safe next step, and identify an appropriate type of eye-care service. It is designed for people who are more comfortable speaking Bengali than using text-heavy English applications.

The product exists to reduce delay, fear and confusion before treatment—not to replace clinicians. It provides a trusted first step when users may otherwise rely on family advice, informal care or forwarded messages.

2. Background and Current State

A rural Bengali-speaking person who notices an eye-health concern may first ask relatives, pharmacists, informal providers or rely on forwarded messages. They may not know whether the concern is urgent, where to travel, what services are available or which advice can be trusted. Text-heavy digital systems can add friction for people with low literacy, limited English confidence, shared devices, weak connectivity or low digital familiarity.

The current journey is therefore fragmented: concern → informal advice → delayed or uncertain decision → travel and service uncertainty. NetraJyoti is intended to create a safer first step before this uncertainty becomes avoidable delay.

3. Problem Statement

Many Bengali-speaking rural users do not know whether an eye problem is serious, where to go for help, or which advice to trust. Cost, travel, waiting time, poor communication and low digital confidence can make formal treatment harder. Delays can worsen avoidable eye-health problems and reduce timely treatment.

4. Evidence and Product Rationale

Evidence signal	Implication for the MVP
Sightsavers Sundarbans research (E1–E2) identifies treatment-seeking, cost, location, informal-first-contact and low-perceived-severity barriers.	Use a short Bengali-first flow, clear red-flag escalation and a referral direction instead of complex medical explanations.
Community Eye Health reporting (E3) describes unequal rural access and geographic difficulty in the Sundarbans.	Show the type of care to seek and support local service discovery; do not assume a hospital is nearby.
Misinformation and usability evidence (E4–E5, E10–E11) highlights trusted forwarded messages and friction in text-heavy systems.	Use concise voice/text prompts, explicit source and safety cues, caregiver sharing and low-reading-load interaction.
Comparator observations (E6–E9) identify waiting, coordination and service-experience concerns.	Set expectations conservatively and ensure service data is verified before public use.

Evidence Reference Key

The labels below identify the specific sources used in this PRD. E1–E5 provide contextual research evidence; E6–E11 are public comparator observations used to frame experience and accessibility risks; E12 identifies an existing West Bengal eye-health support reference. None of these references replaces local user research or clinical review.

Ref.	What the reference is	What it contributes to NetraJyoti

Ref.	What the reference is	What it contributes to NetraJyoti
E1	Sightsavers Sunderbans Endline Survey (2018).	Context on treatment-seeking, financial and service-location barriers; supports referral guidance and the need to reduce uncertainty before travel.
E2	Sightsavers Sunderbans RMP Study (2018).	Context on informal first-contact care and low perceived severity; supports early red-flag prompts and non-judgemental Bengali guidance.
E3	Community Eye Health article: rural eye care in the Sundarbans.	Context on uneven access and geographic difficulty; supports local service-direction and does not assume a nearby hospital.
E4	Harvard Kennedy School Misinformation Review: WhatsApp misinformation in India.	General evidence that trusted messaging channels can spread uncertain information; supports clear sourcing and myth-busting as a future feature.
E5	World Health Organization note on the 'infodemic'.	Public-health context for harmful information overload; supports concise, careful and clearly bounded health communication.
E6	Public Disha Eye Hospitals complaint example.	Comparator signal for patient-experience risk such as coordination or behaviour concerns; supports honest expectation-setting before travel.
E7	Public Disha Eye Hospitals Justdial review page.	Comparator observations about waiting or service dissatisfaction; supports testing of care-journey expectations with local users.
E8	Public Susrut Eye Foundation Justdial review page.	Comparator observations about waiting and support experience; reinforces the need for clear and respectful referral information.
E9	Public Susrut Eye Foundation complaint-style review page.	Comparator observation of service-journey dissatisfaction; supports the decision not to overpromise service quality or availability.
E10	Digital India page describing DigiDrishti.	Comparator context on a multi-layer digital eye-care system; supports keeping the MVP flow short and guided for low-confidence users.
E11	Citizen Eyecare/DigiDrishti public app metadata.	Comparator signal on symptom-checking and language/usability demands; supports Bengali-first voice and text fallback testing.
E12	West Bengal Budget Speech reference to Chokher Alo eye-health support.	Shows relevant state-level eye-health support exists; future service direction should connect users to verified real programmes where appropriate.

Evidence limitation: the desk research supports the hypothesis but does not replace local primary research. Five consented JTBD interviews and partner validation are required before a real pilot.

5. Terminology

Term	Meaning in this PRD
Red flag	A predefined response requiring urgent eye-care guidance, including sudden blurred vision, severe pain, injury or chemical exposure.
Routine guidance	Fixed, non-diagnostic next-step guidance for concerns without configured urgent signals.
Human escalation	Direction to consult an ASHA worker, qualified health worker, clinic or eye-care provider when the app cannot safely classify a concern.
Service directory	A list of eye-care services. MVP entries are demo-only until partner verified.
ASHA worker	Accredited Social Health Activist/community-health worker who may assist a user but does not diagnose through NetraJyoti.
Demo-only data	Prototype information that is displayed only for capstone demonstration. It is not clinically approved and must not be relied on for travel or treatment decisions.

6. Goals, Product Hills and Non-Goals

Goals

- Enable a rural Bengali-speaking user to describe a concern through voice or text without needing English typing.
- Route red-flag responses to immediate urgent-care guidance using fixed safety rules.
- Give non-diagnostic, simple Bengali next-step guidance and an appropriate service type.
- Help caregivers and ASHA/community-health workers support a user without acting as a clinician.
- Create a measurable, clinically reviewable capstone MVP for usability and referral testing.

Product Hills

- A rural Bengali-speaking user can understand the next safe eye-care action without English typing or medical knowledge.
- A caregiver can confidently share the next step without interpreting medical information or making a diagnosis.
- An ASHA/community-health worker can assist a villager through a repeatable flow without being expected to prescribe or diagnose.
- No urgent configured red-flag case is delayed by generated wording, unclear navigation or an unverified service listing.

Non-Goals

- Diagnosing eye disease, interpreting images, prescribing medicines or replacing professional eye care.
- Autonomous appointment booking, medical-record integration or payment processing.
- Publishing unverified provider details, camp dates or emergency availability.
- Using GenAI for triage, diagnosis, prescriptions, free-form clinical chat or altering the Java-selected safe route. The optional local Ollama explanation is post-route only and uses retrieved PostgreSQL guidance.
- Expanding beyond Bengali or rural West Bengal during the MVP stage.

7. Target Users and Jobs-to-be-Done

Persona	Primary job	Key barrier	MVP value
Maya — rural user, 58	Decide what to do when vision changes or the eye is uncomfortable.	Low literacy, fear of cost, English typing and reliance on relatives.	Can speak Bengali and receive short, safe next-step guidance.
Rahim — caregiver, 32	Help a family member decide and prepare for travel.	Uncertain advice and difficulty coordinating care from another location.	Gets a shareable plain-language summary.
Soma — ASHA/community-health worker	Support a villager without giving clinical advice.	Limited time and need for trusted referral guidance.	Uses a guided, non-diagnostic assisted flow with escalation.

Persona	Primary job	Key barrier	MVP value
Outreach coordinator	Keep users aware of appropriate care options.	Outdated service and camp information causes failed referrals.	Maintains partner-verified listings in a later pilot phase.

Persona Profiles and Design Implications

The following provisional personas are grounded in the problem hypothesis and desk research. They must be tested and refined through the required real JTBD interviews; they are not substitutes for primary evidence.

Maya — rural Bengali-speaking user

Context: Maya is an older adult in a rural area who notices a change in vision or eye discomfort. She may use a shared phone, have low confidence reading long text, and depend on relatives for travel or decisions.

- Need: a calm, short answer in Bengali that explains the next safe action without medical jargon.
- Behaviour: may seek family, pharmacy or informal advice before formal care because travel, cost and uncertainty feel high.
- Design requirements: large Bengali labels, one decision at a time, voice input with text fallback, visible urgent warning and no English-only gate.

Rahim — caregiver and decision supporter

Context: Rahim may help a parent or relative from another household or village. He needs enough information to coordinate travel, explain the next step and reduce anxiety without becoming responsible for medical decisions.

- Need: a concise shareable summary that states the urgency and recommended care type.
- Behaviour: compares advice from relatives, local providers and online messages; needs a credible, simple source before asking family to act.
- Design requirements: copy/share action, plain-language outcome, clear disclaimer and service data that is visibly demo-only until verified.

Soma — ASHA or community-health worker

Context: Soma supports several community members with limited time. She needs a repeatable way to help a villager navigate a concern while staying within her non-clinical role.

- Need: an assisted, explainable flow with immediate escalation when uncertainty or red flags occur.
- Behaviour: may complete the flow together with a user and explain the result in familiar language.
- Design requirements: short prompts, fixed safety rules, human-escalation route, no diagnostic label and no expectation that she updates clinical content.

Outreach coordinator — service-information steward

Context: The coordinator works with eye camps, vision centres or hospitals and understands that listing errors can cause avoidable travel and missed care.

- Need: a controlled process to add, update and expire location, contact and service information.
- Behaviour: confirms data with partner organisations rather than relying on user-submitted information.
- Design requirements: future authenticated workflow with source, reviewer, last-reviewed date and expiry. This is explicitly post-MVP; the public MVP uses demo-only entries.

8. Comparative Analysis, Competition and Differentiation

Alternative / competitor	Observed limitation or role	NetraJyoti differentiation
Informal advice: relatives, pharmacies and forwarded messages	Often trusted but may be incomplete, inconsistent or not linked to eye-care services.	Bengali-first safety boundary, fixed next-step guidance and referral direction.
Eye hospitals, camps and vision centres	Provide care but can be difficult to discover or navigate before travel.	Helps the user identify the appropriate type of care before travel; does not claim to replace providers.
DigiDrishti and other text-heavy digital systems	Can require reading confidence, navigation across multiple steps and familiarity with digital health flows.	Rules-based urgency and a Java-controlled route. Optional local Ollama explains only the already selected route from PostgreSQL guidance and is validated by Java; it cannot diagnose, prescribe, alter urgency or invent service details.
General AI/chat tools	May produce unverified or diagnostic-looking health answers.	Rules-based urgency and fixed clinician-reviewable wording. A controlled pilot may simplify only server-approved copy; it never generates clinical advice.

Competitive position: NetraJyoti is not a hospital or diagnostic platform. Its differentiated role is a trustworthy, Bengali-first first step between informal advice and verified formal eye care.

9. MVP Scope and Feature Catalogue

Feature catalogue

Priority	Features
Must have	Bengali voice input with text fallback; acknowledgement; Java deterministic routing using PostgreSQL criteria; fixed route action cards; optional local Ollama RAG explanation; demo-only service direction; browser share/copy family message.
Controlled pilot only	Clinical approval of the PostgreSQL guidance corpus; AI-assisted labelling; feedback capture; partner-verified directory management; ASHA-assisted workflow; public-pilot security and governance controls.
Should have	Feedback capture; ASHA-assisted workflow; verified-directory management; eye-camp reminders after partner validation.

Priority	Features
Could have	Offline content cache; WhatsApp sharing; partner-approved appointment-request support; clinician-approved FAQ knowledge base.
Won't have in MVP	Disease or image diagnosis; prescriptions; autonomous booking; medical-record integration; publicly relied-upon unverified provider data.

In scope	Deferred / out of scope
Bengali voice input with text fallback; fixed-category routing; Java fixed route action cards; optional local Ollama RAG explanation from PostgreSQL route guidance; static demo service direction; browser-native share/copy family message; safety/privacy acknowledgement.	Text-to-speech result audio; clinical approval of content; persistent feedback; verified public directory; automated booking, WhatsApp integration, payment or EHR connectivity.
Rules-based urgency checks for sudden vision change, severe pain, injury and chemical exposure. Java/PostgreSQL routing determines the result before any Ollama request.	Prescriptions, treatment plans or dosage advice.
Fixed Bengali Java fallback guidance for urgent, routine and human-support routes; optional Java-validated local Ollama explanation of the already selected route.	Generative AI health answers, free-form clinical chat, diagnosis, prescriptions or AI-driven triage.
Static demo service direction and browser share/copy family message. The UI does not persist feedback and does not provide result-page audio.	Verified public directory, automated booking, WhatsApp integration, payment or EHR connectivity.

10. Required User Stories and Acceptance Criteria

ID	User story	Acceptance criteria
US-01	As a rural Bengali-speaking user, I want to describe my eye problem by voice so I do not need to type or read English.	The user can start Bengali speech input; recognised text appears for review; text entry remains available if voice input fails; the app states that voice capability depends on the device/browser.
US-02	As a user with sudden vision loss, severe pain, eye injury or chemical exposure, I want immediate advice to seek emergency care.	Any configured red flag returns an URGENT result; the screen and audio say not to wait; no diagnosis or medication recommendation is shown; relevant care types are suggested.
US-03	As a user with a non-urgent concern, I want simple Bengali information on what to do next without being told I have a disease.	Routine cases receive fixed, simple Bengali guidance; the disclaimer is visible; response language contains no diagnosis, prescription or unsupported disease claim.
US-04	As a caregiver, I want a short referral summary so I can help my family member travel and prepare.	A Bengali summary can be copied or shared using supported device sharing; it includes the next step and disclaimer but no diagnosis.
US-05	As an ASHA worker, I want an assisted flow so I can guide a villager safely without acting as a clinician.	The same guided flow can be completed together; uncertainty routes to human escalation; the interface consistently states that it is not diagnosis.

ID	User story	Acceptance criteria
US-06	As an outreach coordinator, I want to update eye-camp dates and locations so users receive current service information.	Not in the public MVP UI. A future authenticated partner workflow must record source, reviewer, last-reviewed date and expiry; unverified entries must never be marked verified.
US-07	As a Bengali-speaking user, I want an approved next-step message explained in simpler Bengali so I can understand it easily.	Current live implementation: Java selects the route before the Ollama call; the backend sends only retrieved Bengali route-guidance content to local Ollama, not free text, voice, location or personal details. Java validates output and falls back to fixed Bengali text. Gap before a controlled pilot: use clinically approved guidance records and clearly label accepted output as AI-assisted.
US-08	As a caregiver, I want a short plain-Bengali summary of an approved result so I can help my family member act on the next step.	Current live implementation: a user can review, copy or share a family message containing the selected route guidance and symptoms. It is not a separate AI caregiver-summary capability derived solely from approved final-result content. That capability remains controlled-pilot work.
US-09	As a NetraJyoti user, I want clear consent and privacy controls so I understand how my information is handled.	Current live implementation: acknowledgement is required before assessment. Optional patient details remain in browser state and are included in a shared message only after a separate checkbox acknowledgement. Research, analytics and recording consent are deferred.
US-10	As the product team, we want safety, accessibility and fallback checks so the MVP remains dependable for Bengali-speaking users.	Current live implementation: routing-policy unit tests and Bengali text fallback exist. End-to-end device, voice, slow-network, directory and usability testing remain release-gate work.
US-11	As the product team, we want privacy-safe pilot feedback and analytics so we can measure usability without unnecessary data collection.	Deferred. No feedback or analytics UI, API or persistent store is present in the current live MVP.

11. Functional Requirements

ID	Area	Requirement
FR-01	Start and consent	Show the non-diagnosis and emergency disclaimer before the assessment; require acknowledgement of the privacy statement before continuing.
FR-02	Concern capture	Accept only approved concern categories: blurred vision, pain, redness/watering, injury, chemical exposure or other. Reject invalid API values.

ID	Area	Requirement
FR-03	Safety rules	Return URGENT for configured injury/chemical exposure, severe pain or sudden vision change. Return HUMAN SUPPORT for Other/unclear concerns, explicit request for a person, escalation criteria or no eligible demo criterion. Otherwise return ROUTINE. Java applies fixed precedence: URGENT, HUMAN SUPPORT, ROUTINE.
FR-04	Guidance output	The live MVP shows fixed Bengali result text and a Java-controlled action card. Browser speech recognition is offered as optional input with a typed fallback; result-page audio is not provided.
FR-05	Service direction	Recommend a type of care based on the selected route. The live UI displays static demo service options and asks the user to verify locally before travel. The PostgreSQL service-location repository is not yet connected to the UI.
FR-06	Caregiver summary	Allow a concise, shareable family message containing the selected route guidance and symptoms; use browser-native sharing or copy as fallback. Optional patient details remain on-device and are added only after a separate acknowledgement.
FR-07	Feedback	Deferred. The live MVP does not capture a clarity/usefulness rating, optional comment, or feedback acknowledgement.
FR-08	Controlled GenAI simplification	Live MVP: after Java selects the route, an explicit backend flag may send only PostgreSQL-retrieved Bengali route-guidance content to local Ollama. It cannot change the route; Java validates output and uses fixed Bengali fallback on error, unavailable retrieval or unsafe output. The current retrieved records are demo-simulated, not clinically approved. AI-assisted labelling and clinically approved content are required before a controlled pilot.
FR-09	Controlled AI-assisted caregiver summary	Deferred controlled-pilot capability. The live MVP provides a browser share/copy family message, not a separate AI-generated caregiver summary from approved result content.

Requirements Traceability

PRD requirement	Jira delivery item	Status
FR-08 — Controlled GenAI simplification	NET-11 — AI-assisted plain-Bengali explanation of approved guidance	Controlled-pilot story
FR-09 — AI-assisted caregiver summary	NET-12 — AI-assisted caregiver summary from approved result content	Controlled-pilot story (3 points)

12. Safety, Privacy and Ethical Requirements

Safety requirement: Java deterministic routing is the only component that selects the route. It retrieves eligible PostgreSQL criteria and applies fixed precedence: URGENT, then HUMAN SUPPORT, then ROUTINE. Optional local Ollama runs only after the route is fixed and cannot change urgency, diagnose, prescribe or add service details. Demo-simulated data are not clinically approved and are not production eligible.

- Display 'Safe information, not a diagnosis' and urgent-care instructions throughout the flow. The route-result action card is Java controlled.
- The live routing API receives selected fixed concern codes, urgent-sign codes, optional structured history and an explicit request for human support. Free text, browser speech transcripts, location and patient details do not determine the route.
- Optional free text and voice transcripts remain in the browser and are not sent to the routing or Ollama API. Optional patient details remain in browser state and are included only in the user-selected shared message after a separate acknowledgement.
- The live Ollama request contains PostgreSQL-retrieved Bengali guidance and the Java-selected route. It does not contain free text, voice transcript, location, patient details or identifiers. Current route-guidance records are explicitly demo-simulated, not clinically approved.
- Ollama is configured server-side and is local by default. Java validates every candidate response and returns fixed Bengali fallback copy when Ollama, retrieval or validation fails. The current UI does not explicitly label accepted output as AI-assisted; add that label before a controlled pilot.
- The live family-message feature is browser share/copy of the selected route guidance and selected symptoms. A separately generated AI caregiver summary from clinically approved result content is deferred.
- Java rejects unsafe or route-inconsistent Ollama output, including diagnosis, medicine/dosage, treatment, cure, changed urgency or care type. It shows fixed Bengali fallback copy instead.
- The live UI uses static demo service options. A PostgreSQL service-location repository exists but is not integrated into the UI. Partner verification, source, reviewer, last-reviewed date and expiry are required before public use.
- Human support is available when the user explicitly requests help, selects Other/unclear concern, criteria escalate, or no eligible demo criterion is available. An urgent outcome always takes precedence.
- Current demo logging includes route input codes in backend logs. Before a pilot, change this to non-content audit metadata only: selected route, rule version, feature-flag state, validation result and fallback reason. Do not retain health-selection codes, location or identity.

Controlled-GenAI Operational Ownership

Responsibility	Named role / requirement
Feature-flag and provider-key owner	Backend owner; enables the pilot only after release gates and keeps the key in server-side configuration.

Responsibility	Named role / requirement
Approved Bengali source-content owner	Qualified clinical/content reviewer; owns approved messages, caregiver templates, review date and expiry cadence.
Output-evaluation owner	Product and clinical reviewers; retain non-content evaluation evidence for faithfulness, prohibited-content rejection and fallback behaviour.
Incident owner	Pilot operations owner; disables the feature flag, preserves the fixed-copy path and routes safety concerns to the defined escalation process.
Technical logging policy	Live demo status: selected concern/history codes and matched criteria are currently logged. Pilot requirement: replace this with non-content audit metadata only - route, rule version, feature-flag state, validation result and fallback reason - and do not log health selections, location or identity.

13. Non-Functional Requirements

Area	MVP requirement
Accessibility	Mobile-first layout; Bengali-first labels; touch targets; browser speech input and typed alternatives; readable contrast; no mandatory English input.
Performance	The live UI shows the Java-selected route and fixed action card immediately, while the optional Ollama explanation loads asynchronously. The current Ollama read timeout is 75 seconds; a <=10-second AI response/fallback objective remains a pilot release gate.
Reliability	If voice input, sharing or service lookup fails, keep text guidance visible and present a practical fallback.
Security	HTTPS in deployment; API input validation; server-side configuration only; no secret keys in the browser; rate limiting before public pilot.
Controlled GenAI	Live MVP: feature flags default off; when enabled, local Ollama receives PostgreSQL-retrieved Bengali route guidance after Java selects the route. Java validates bounded output and returns fixed Bengali fallback. Clinical approval of the corpus and explicit AI-assisted labelling are required before a controlled pilot.
Maintainability	Rules, Bengali wording and service listings must be versioned and reviewable; safety changes require clinician approval.

14. User Flow

1. User opens NetraJyoti and sees the Bengali-first purpose, urgent warning and non-diagnosis boundary.
2. User acknowledges the safety/privacy statement before the assessment begins.

3. User selects one or more fixed symptoms, checks urgent warning signs, may select optional Step 3A history, and may speak or type optional detail. Voice/free text do not determine the route.
4. Java retrieves matching PostgreSQL criteria and deterministically selects URGENT, ROUTINE or HUMAN SUPPORT. Urgent has highest precedence. In the current demo, Step 3A history is displayed as context and the simulated seed data contain no history criteria that alter the route.
5. The user sees the fixed route action card and selected symptoms. The UI then collects district/town or location only for demo service direction; this does not change the route. Result-page audio is not part of the live MVP.
6. When explicitly enabled, PostgreSQL retrieves Bengali guidance for the Java-selected route and local Ollama generates one concise Bengali explanation. Java validates it; on failure, the fixed Bengali fallback remains visible. The current source records are demo-simulated, not clinically approved.
7. User can view static demo service direction, review/copy/share a family message, go back, return to the current route result or restart. Feedback capture is deferred.

14.1 End-to-End User Journey Map

Purpose. The journey below makes the safe routing logic visible from the user perspective. It describes what users do, what NetraJyoti does, and the next screen or action. This is a navigation journey, not a diagnostic workflow.

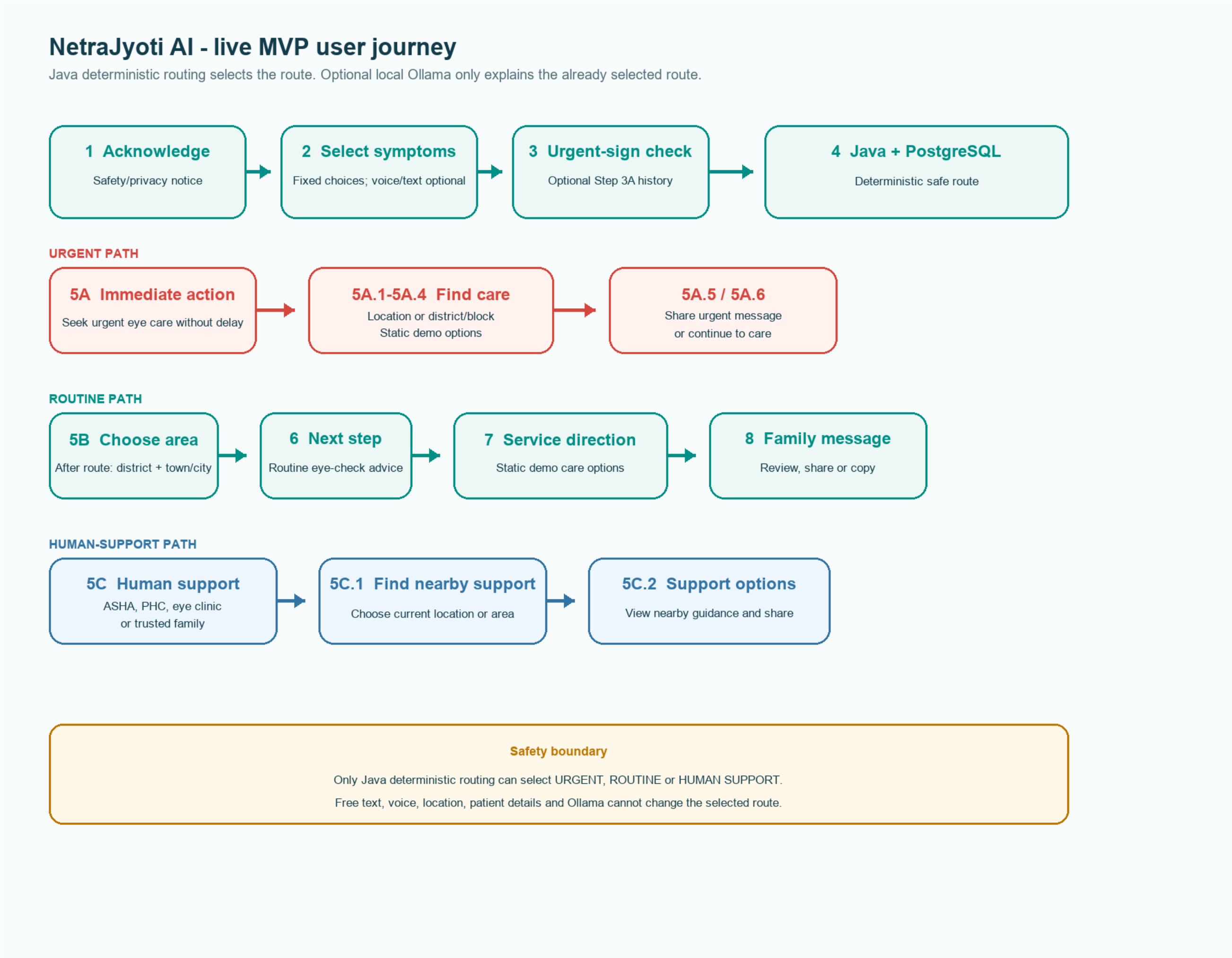

Figure 1. NetraJyoti AI end-to-end user journey and safe-route branches.

Stage	User experience	NetraJyoti response	Next
1. Welcome and consent	Reads Bengali-first purpose, warning notice, and consent statement; selects consent.	Enables “Start safely”. No personal details are requested.	Step 2
2. Select symptoms	Selects one or more fixed eye symptoms. Voice/text is optional.	Captures approved IDs. Optional detail never decides urgency.	Step 3
3. Urgent-sign check	Answers whether any urgent warning sign is happening now.	Records fixed urgent-sign IDs for reviewed rules.	Step 4
4. Safe guidance check	Reviews confirmation that details and warning signs were checked.	Rules select URGENT, ROUTINE, or HUMAN SUPPORT. No disease label is shown.	5A / 5B / 5C
5A. Urgent action	Receives immediate-care advice; can find demo care options, share, or continue to care.	Keeps the Java-controlled urgent action visible. Optional Ollama explanation loads asynchronously and cannot change the route.	5A.1–5A.6
5B–8. Routine route	Chooses district/town after the Java route is selected, reviews	Shows the Java-controlled routine action card immediately. Optional Ollama explanation loads	Complete / restart

	recommendation/service direction, and prepares a family share message.	asynchronously and cannot change the route. Static demo service options are shown on the next screen.	
5C–5C.2. Human support	Chooses ASHA, PHC, eye clinic, family, or nearby-support options.	Provides a human-support route when concern is unclear or support is requested.	Support / share / restart

Journey principle. At every stage, the user can understand the next action and has a safe option to go back, return to the current route result, or start again. Location, voice, sharing, and controlled AI are optional enhancements; their absence must not block safety guidance.

14.2 Transition and Branching Rules

The following transitions align the PRD with the live MVP and detailed wireframe. The trigger states the specific user action or reviewed rule that moves the journey forward.

Transition	Trigger / decision	Route
1 → 2	Consent selected and “Start safely” tapped	Core
2 → 3	One or more symptoms selected and 'Check urgent warning signs' tapped	Core
3 → 4	Urgent-sign check completed; optional Step 3A history selected or skipped; then 'View safe guidance' tapped	Core
4 → 5A	Java/PostgreSQL criteria detect an urgent warning sign	Urgent
4 → 5B	Java/PostgreSQL criteria select ROUTINE; district/town is collected after the route for service direction only	Routine
4 → 5C	Concern is unclear, “other” is selected, or human support is requested	Human support
5A → 5A.1	“Find verified care” tapped	Urgent
5A.1 → 5A.2 → 5A.4	Current location selected, permission granted, and nearby services returned	Urgent
5A.1/5A.2 → 5A.3 → 5A.4	District/block or PHC selected; then verified services requested	Urgent
5A → 5A.5	“Share urgent message” tapped	Urgent
5A → 5A.6	“Continue to care” tapped	Urgent
5B → 6 → 7 → 8	Area selected → safe guidance → service direction → family message	Routine
5C → 5C.1 → 5C.2	Nearby support → area/current location → nearby support viewed	Human support

Safety boundary. Only the reviewed rules module determines the urgent, routine, or human-support route. Free text, voice input, location, patient details, service selection, and AI output must never change urgency.

15. Success Metrics and Pilot Validation

Metric	Pilot measure	Initial success threshold
Flow completion	Percentage of consented users completing concern capture and receiving a result.	≥70% with assisted onboarding documented separately.
Red-flag routing	Percentage of scripted red-flag test cases routed to URGENT.	100% in test suite and clinical review.
Comprehension	User can repeat their next step in their own words after task completion.	≥80% of usability-test participants.

Metric	Pilot measure	Initial success threshold
Voice usability	Users who can successfully use browser speech recognition or switch to typed input without abandoning the flow.	≥75%; segment by device/noise/assistance.
Directory accuracy	Partner-confirmed availability and contact details.	100% before any non-demo listing is shown.
Trust and safety	Users reporting that guidance was clear; unresolved/unsafe sessions flagged for review.	No unresolved urgent test case; qualitative issues logged.
Caregiver-summary comprehension	Deferred controlled-pilot measure. The live MVP has a share/copy family message, not a separate AI caregiver summary.	≥80% in controlled-pilot usability sessions.
Outcome fidelity	Sampled validated Ollama explanations preserve the Java-selected route, urgency and next-step boundary.	100% in evaluated test corpus.
Safety fallback	Rate of failed/rejected Ollama requests that display fixed Bengali fallback; verify that free text, voice, location and patient details never reach Ollama.	100% safe fallback; 0 user-data transfers in test evidence.

16. Testing and Release Gates

8. Run unit tests for all red-flag, routine and human-escalation paths; reject invalid concern values.
9. Conduct clinical review of safety rules, Bengali scripts and disclaimers with an ophthalmologist or authorised eye-care partner.
10. Conduct task-based usability sessions with rural users, caregivers and ASHA/community-health workers; test comprehension, voice recognition and fallback text entry.
11. Validate every service entry, contact route, hours and camp date with a partner before moving from DEMO ONLY to pilot data.
12. Test slow networks, older devices, microphone denial, browser voice unavailability, empty location and service lookup failure.
13. Complete privacy review, deploy HTTPS, protect configuration and add operational monitoring/rate limiting before any public pilot.
14. Before enabling the controlled GenAI pilot, obtain clinician sign-off on source messages, privacy approval, prompt/output evaluation evidence, explicit prohibited-content validation tests, non-content audit logging checks and a tested fallback to the fixed Bengali message and caregiver summary.

17. Dependencies, Risks and Decisions

Dependency / risk	Mitigation or decision

Dependency / risk	Mitigation or decision
Clinical safety review is not yet evidenced as complete.	Do not launch publicly. The current PostgreSQL pathways and Bengali guidance are marked DEMO_SIMULATED_NOT_FOR_CLINICAL_USE; obtain documented clinical review, approval, versioning and ownership before a pilot.
Service data may be stale or unavailable.	Keep demo-only warning now; add authenticated, partner-owned update and expiry workflow later.
Speech quality varies by device, accent and noise.	Use browser speech only as an assistive input; provide text fallback and test with target users.
Local research is incomplete.	Conduct five real, consented JTBD interviews; retain raw notes, exact quotes and recording references; do not fabricate findings.
Users may interpret guidance as diagnosis.	Repeat boundary language at consent, result and share points; avoid disease names and treatment recommendations. The live UI meets this boundary, but usability validation remains required.
GenAI simplification could introduce an unsafe or misleading statement.	The live local Ollama path is post-route and Java-validated. Before a pilot, use clinically approved source content, add explicit AI-assisted labelling, bound response time and retain non-content evaluation evidence.
A local Ollama service may be unavailable or inadvertently exposed.	Keep local Ollama behind the backend and do not expose it to the browser or public internet; retain Java fallback when unavailable.
Current demo logs selected health codes and matched criteria.	Before pilot, redact/stop content logging and retain only non-content audit metadata. Validate this with automated log tests.
Step 3A history is accepted by the repository routing-query contract even though the current demo UI describes it as context only.	Keep simulated and future approved history criteria out of route selection unless a revised PRD, clinical review and explicit user-facing explanation authorise that change.

18. Future Roadmap (Post-MVP)

- Authenticated coordinator tools for verified service, camp and referral updates.
- Clinician-approved Bengali knowledge base, retrieval controls and safety evaluation for limited non-diagnostic FAQs.
- Expand the controlled AI-assisted plain-language pilot only after documented comprehension, privacy, safety and fallback results; do not broaden it into clinical chat without a separate PRD and governance review.
- ASHA-assisted mode, reminder workflow and an offline content cache following field validation.

- Partner-approved referral follow-up and aggregate analytics with consent and minimum-data safeguards.
- Broader district coverage only after local evidence, service partnerships and safety governance are established.

Next-Phase GenAI Capability Candidates

The following are product opportunities, not committed MVP functionality. Each requires a separate backlog story, clinician-approved source content, privacy review, output evaluation, feature flag, human-escalation route and safe fixed-copy fallback before pilot use.

- Explain clinician-approved eye-health FAQs in simpler Bengali without adding clinical facts.
- Provide voice-led clarification of approved referral and visit-preparation instructions.
- Create a concise Bengali caregiver summary only from the already-approved result content.
- Help an ASHA/community-health worker navigate approved, non-diagnostic guidance during an assisted session.
- Recognise that a request is unclear or outside the approved content and direct the user to a qualified health worker instead of attempting an answer.

Explicit boundary: These next-phase candidates must never diagnose, interpret images, prescribe, classify urgency, replace the reviewed rules engine, or generate ungrounded health advice.

19. Appendix

Appendix	Contents
A	Evidence register E1–E12 and source links in NETRAJYOTI.docx.
B	JTBD interview script, consent wording, raw-notes template and recording inventory in the Phase 2 submission pack.
C	Safety escalation matrix, clinical-review gate and service-data verification requirements.
D	NetraJyoti MVP wireframe, user flow and Java/React technical implementation.
E	Test cases, pilot metrics and release-gate checklist.

20. Source Basis

This PRD synthesises the project summary and evidence appendix in NETRAJYOTI.docx and the Phase 2 strategy, interview plan, safety matrix, MVP plan and user stories in NetraJyoti_Capstone_Project_Submission_Pack_v2.docx. Version 5.2 also records an implementation review of the current React/Vite UI, Java/Spring Boot API, PostgreSQL seed data and local Ollama integration as of 3 September 2026. Evidence references E1-E12 and comparator materials remain in the source documents. This PRD does not claim that the required five real interviews or clinical validation have already been completed.